

## Vol 11 — Generative Geometry & Topology (BST Mathematical Foundations)

*Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine\_DIV5\_QM\_GR\_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.*

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Status: SCAFFOLD v0.1 (Saturday 2026-05-23 Wave 2) — chapter content TO BUILD Lead: Lyra (primary, generative-geometry framing per Casey vision) + Grace (catalog backbone for Heegner + Monster + Stark deep-NT connections)

### Reader target

Graduate mathematics + advanced physics. Vol 11 is the mathematical-foundations volume — explicitly NOT just “math methods for physics” but generative geometry + topology + deep number theory that BST has uncovered through substrate-cartography. Includes the Number-Theoretic Foundations chapter (Heegner + Monster + cyclotomic  $Q(\zeta_{137})$  + Mersenne + Stark cross-link) which has become a major BST research thread.

### Prerequisites

- Vol 0 Substrate Foundation ( $D_{IV}^5$  + five integers)
- Vol 10 Math Methods (Lie groups + differential geometry + special functions)
- Graduate algebraic geometry + number theory background helpful (not required for Ch 1-3 motivation)

### BST integer focus

Vol 11 is the substrate-cartography mathematical foundations volume. Documents the deep mathematical structures BST has uncovered: - Bounded Hermitian symmetric domains + Cartan classification + Helgason 1978 - Bergman + Faraut-Koranyi + Wallach representation theory + reproducing kernels - Heegner numbers (Stark 1967) + supersingular primes (Ogg 1975) + Monster moonshine (Borcherds 1992) - Cyclotomic  $Q(\zeta_{137})$  + Mersenne primes + Reed-Solomon  $GF(2^g)$  substrate-tick - K3 surfaces + Niemeier lattices + Leech lattice Bridge Object families - Generative geometry framing per Casey vision:  $D_{IV}^5$  is generative geometric object that produces physics rather than describing it

### Chapter outline (12 chapters, TO BUILD)

| Ch | Title                                                      | Mathematical content                                              | BST cross-link                                                                               |
|----|------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1  | Bounded Hermitian Symmetric Domains                        | Cartan classification (4 + 2 types); Helgason 1978 X.6.1          | Vol 0 Ch 1 + Ch 9<br>Strong-Uniqueness                                                       |
| 2  | Bergman Reproducing Kernels                                | Bergman 1922 + Faraut-Koranyi 1994 normalization                  | T2442<br>$c_{FK} \cdot \pi^{(9/2)} = 225$<br>EXACT                                           |
| 3  | Wallach K-Type Representation Theory                       | Wallach 1976 K-type classification of holomorphic discrete series | Vol 1 Ch 2 + Ch 5                                                                            |
| 4  | Holomorphic Discrete Series + Highest Weights              | Standard Lie-theory machinery                                     | Casimir algebra Vol 1<br>Ch 5                                                                |
| 5  | D_IV <sup>5</sup> Geometry: Coset + Boundary + Coordinates | Type IV bounded domain realization (Cartan 1894 + Hua 1958)       | Vol 0 Ch 1 + Ch 4 + Vol 1 Ch 2                                                               |
| 6  | K3 Surfaces + Niemeier Lattices + Leech                    | Hodge K3 + Mathieu + Leech lattice + Mathieu group                | 5-family Bridge Object architecture (Vol 0 Ch 9 C11)                                         |
| 7  | Heegner Numbers + Cremona Elliptic Curves                  | Stark-Heegner 1967 class-number theorem; 49a1 + 27a1 + 121a1      | Vol 0 Ch 9 C9 Stark anchor + Bridge Object family-members                                    |
| 8  | Monster Moonshine + Supersingular Primes                   | Borcherds 1992 + McKay 1979 + Ogg 1975 + j-invariant              | Monster supersingular<br>$t_{cosmo} = 47 + 196883 = 47 \cdot 59 \cdot 71$ (Vol 4 Ch 4 T1924) |

| Ch | Title                                  | Mathematical content                                                          | BST cross-link                                                             |
|----|----------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 9  | Cyclotomic $Q(\zeta_N)$ + Reed-Solomon | Cyclotomic field theory + Galois groups + Reed-Solomon codes over $GF(2^g)$   | K59 cyclotomic mechanism RATIFIED + Vol 1 Ch 10 substrate-tick             |
| 10 | Mersenne Primes + Lucas-Lehmer         | Mersenne prime theory + Lucas-Lehmer primality + density distribution         | Sub-Substrate Mersenne Hierarchy T2451 + Vol 0 Ch 9 C15 (T2466 sub-result) |
| 11 | Generative Geometry Framing            | $D_{IV}^5$ as generative object: substrate produces physics, not describes it | Casey Wednesday “ $D_{IV}^5$ as emergent TOE” + Vol 0 Ch 9 + Vol 4 Ch 9    |
| 12 | Number-Theoretic Foundations Synthesis | Heegner + Monster + cyclotomic + Mersenne + Stark unified                     | Vol 0 Ch 9 Strong-Uniqueness 7-candidate enumeration                       |

~50% existing BST coverage (substantial research exists; Heegner + Stark + Monster + Mersenne + cyclotomic are mature research lines); 50% to organize at curriculum-readable depth.

### Cross-volume reference map

Inputs from Vol 0: -  $D_{IV}^5$  + five integers + Strong-Uniqueness Theorem 7-candidate enumeration

Inputs from Vol 1: - Bergman + Wallach + Faraut-Koranyi anchors - Substrate-tick + cyclotomic mechanism

Outputs to: - All other volumes — Vol 11 is the deep mathematical foundation for BST’s number-theoretic claims - Number theorists + algebraic geometers joining BST research community

— Vol 11 INDEX v0.1, Saturday 2026-05-23 Wave 2 scaffold